

EIS-Based State of Charge Characterization of Electric Double-Layer Capacitors

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Abstract—This paper presents an experimental study performed on the state of charge (SoC) characterization of two commercial electric double-layer capacitors (EDLCs), rated 60 F and 400 F, using electrochemical impedance spectroscopy (EIS). A custom-designed four-terminal fixture was designed to improve the measurement accuracy. The data quality was assessed with a Kramers-Kronig analysis and by evaluating Lissajous plots and open-circuit potential stability. A Bisquert open-circuit equivalent circuit model was fitted to the measured impedance spectra, capturing parameters such as electrolyte resistance, charge transfer resistance, pore length, the equivalent series resistance (ESR), and capacitance at different SoC levels. The fit captures the measurement data with an average relative RMSE below 5%. The results show that the ESR and pore length decrease as the SoC decreases; capacitive dominance is stronger at lower SoC. The approach presented in this paper yields an accurate model that provides insight into the physical meaning behind the parameters, which allows for refined SoC estimation and broader EDLC system optimization.

Index Terms—Electric double-layer capacitors (EDLC), electrochemical impedance spectroscopy (EIS), equivalent circuit models (ECM), supercapacitor (SC), and state of charge (SoC).

I. INTRODUCTION

Supercapacitors (SCs) are slowly becoming indispensable in modern power systems, complementing batteries by providing far superior performance in areas where batteries are lacking: their high power density, exceptionally long cycle life, and much wider operational temperature ranges [1]. Over the past three decades the active-material energy density of SCs has improved by almost twenty-fold, now reaching an electrode-level 39 Wh kg^{-1} [2] [3]. This has broadened the applications of SCs from grid-frequency smoothing and renewable-energy buffering, through regenerative braking storage in electric/hybrid public transportation to jump starting trucks in Siberia and being a viable energy storage in LEO satellites [4]. As more critical fields start incorporating SCs, it is imperative to characterize, test, and develop SCs. We know from battery research that many effects, such as ageing, capacitance

degradation, power fluctuations during operational usage, and physical phenomena, can be identified or detected in an EES devices by measuring the alteration in the impedance's frequency response. This is where Electrochemical Impedance Spectroscopy (EIS) comes in [5].

EIS allows for the deconvolution of different electrochemical phenomena that occur on different time scales or, in other words, with different characteristics and frequencies; processes that take place at the same time but occur on different timescales [6], [7]. In supercapacitors, EIS measurements can quantify the equivalent series resistance (ESR), electrolyte resistance, pore length, frequency-dependent capacitance, and ion diffusion effects, providing insight into how these properties vary with SoC. However, accurately modelling supercapacitor impedance requires accounting for the porous electrode structure [8]. EIS modeling is well established. Foundational studies have demonstrated efficacy of using a transmission line to predict the dynamic voltage response of EDLCs in demanding applications. However, while effective for system-level simulation, models based on a composite time constant do not fully deconstruct the distinct physical phenomena occurring within the electrode pores [9]. This limitation is critical, as recent work on Hybrid Supercapacitors (HSCs) highlights a growing community need for more granular characterization—calling for public datasets and testing standards that go beyond single-frequency manufacturer values. Their experimental work revealed a strong temperature dependency in HSCs, creating a need to apply similarly rigorous methods to the more temperature-resilient EDLC technology [10].

This work fills that gap. We perform a detailed EIS characterization of 60 F and 400 F EDLCs across a full range of SoC levels. The resulting impedance spectra are fitted using the Bisquert open-circuit model to extract key equivalent circuit model (ECM) parameters. This approach reveals how parameters such as distributed capacitance and characteristic time constants evolve with SoC and between devices of

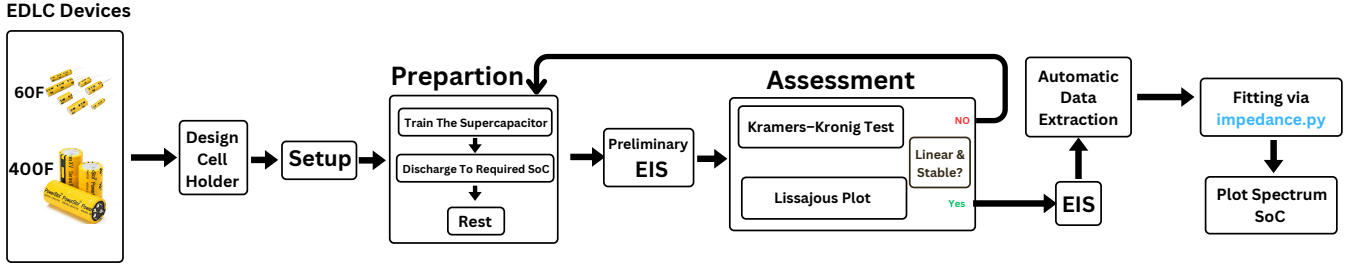


Fig. 1: Methodology procedure for EDLC devices of 60 F, and 400 F. Key steps include initial measurement validation (Kramers–Kronig analysis and Lissajous plots to ensure a linear, stable system), a preliminary EIS run and test-cell optimization, SC conditioning (training cycles and rest), setting the desired SoC, and automated impedance data extraction.

different sizes. The objectives are to validate the model’s effectiveness in capturing SoC-related impedance variations thus providing insights critical to refined SoC estimation, simplified model calibration, and enhanced supercapacitor management strategies in dynamic applications.

The structure of this paper is as follows. Section II presents the experimental methodology, including device preparation, experimental setup, validation procedures and equivalent circuit modelling approach. Section III presents the experimental results, highlighting SoC-dependent impedance spectra and model fitting outcomes. Finally, the conclusions and indications for future work are given in Section IV.

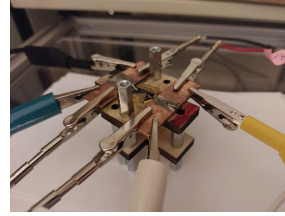
II. METHODOLOGY

This study aims to model the SoC of two different EDLC devices with different storage capacities based on EIS data. Our methodology starts with building the test holders, configuring the SC test schedules and rest times, extracting EIS data, and fitting the ECMs. Fig. 1 shows a flow chart illustrating the EIS measurement procedure for the EDLC devices tested and the entire research methodology.

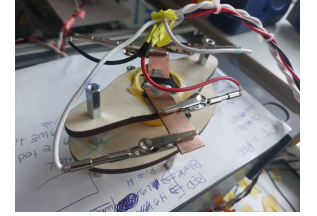
The EDLC devices are as follows:

- 60 F EDLC (EATON TV1840-3R0606-R): A medium-sized device suitable for quick discharge of charge.
- 400 F EDLC (EATON XV3560-2R7407-R): A larger capacity snap-in pin-styled capacitor designed for PCB-based capacitor banks mounted.

Unlike conventional lithium-ion cells, commercially available holders for supercapacitors are limited. Therefore, custom laser-cut fixtures were designed and fabricated for the 60 F and 400 F cells to ensure secure and consistent testing conditions. Proper four-terminal (Kelvin) connections were implemented using copper busbars for the two supercapacitors, effectively separating the current-carrying leads from the voltage-sensing leads. This ensures repeatable EIS measurements at low frequencies. In a general EDLC device, the dominant ohmic drop primarily arises from electrolyte resistance, followed by contact resistance. However, in larger SCs (60 F and 400 F), electrolyte design has been optimized to reduce interparticle and electrolyte resistance, making contact resistance the more significant factor. The busbars employed in our tests mitigate this. Ideally, these connections should be welded for minimal



(a) 60F holder with insulated Arbin and bare EIS clips.



(b) 400F holder featuring busbars for EIS connections.

Fig. 2: Customized SC holders designed to ensure stable and repeatable connections during EIS measurement.

resistance; however, since welding copper to copper is inherently challenging, the busbar was well-soldered instead.

A. Device Preparation and Preconditioning

Before the EIS measurements, each SC was preconditioned to reach a stable state (minimizing leakage current due to self-discharge). This is important as it allows for more accurate SoC fits, as the charge capacity does not change during the EIS measurements.

1) *Prolonged Charge Duration*: Increasing the charge duration will decrease the self-discharge mechanism. Self-discharge can be classified approximately as a function of charge history and temperature. For an EDLC SC, it is caused by internal impurities that cause redox reactions. These impurities are the reaction groups that are distributed very sparingly in active carbon material [11]. This is important since if a constant potential is applied, the SC can be forced to continuously drive the redox reaction in a manner that depletes the reactive species. The reactive species involved in self-discharge are either converted into stable products or become less available due to diffusion limitations leading to a decrease in leakage current over time [12].

2) *Training*: Each EDLC was cycled multiple times to reach a quasi-equilibrium state. Repeated charge–discharge cycles “train” a SC by saturating the slow processes that contribute to self-discharge [13]. The use of both preconditioning methods, voltage hold and SoC cycling, ensured each supercapacitor reached a stable condition before EIS testing.

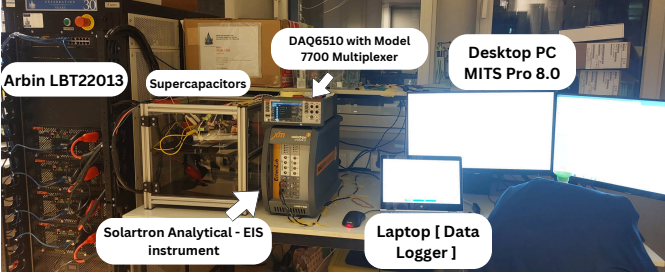


Fig. 3: Experimental setup for EIS measurements on EDLC supercapacitors. The setup includes an Arbin LBT22013 battery tester for charge/discharge control, a Solartron Analytical impedance analyzer for frequency-domain measurements, and a Keithley 7700 multiplexer for automated channel switching.

B. Experimental Setup

The EIS experiments were conducted at room temperature using an Arbin LBT22013 battery tester (for charging/discharging) and a Solartron Analytical impedance analyzer (for EIS measurements) connected via a Keithley 7700 multiplexer. The role of the multiplexer is to connect the battery tester and Solartron to change between SCs quickly. The general test setup is shown in Fig. 3. All tests were performed in an open circuit after the cells were conditioned to a target SoC. Since state of charge (SOC) is linearly proportional to the open-circuit potential (OCP) of the EDLC, the SOC can be defined as a percentage of the current OCP (OCP_c) at any moment of time versus the working voltage (V_t) specified in each supercapacitor's datasheet.

$$SOC = \frac{OCP_c}{V_t} \times 100 \quad (1)$$

Each device was allowed to rest at the desired SoC (after charging) until the terminal voltages reached equilibrium.

1) *Measurement Procedure:* EIS was performed in galvanostatic mode using a small AC current excitation, and the resulting voltage response was measured over a broad frequency range. Galvanostatic EIS is preferred for supercapacitors because of their extremely low internal impedance, making potentiostatic control difficult.

The AC current amplitude was chosen to keep voltage perturbations within the linear regime. The frequency was swept from high to low on a logarithmic scale. Table I summarizes the key EIS parameters for each device.

TABLE I: EIS measurement parameters

SoC	60F	400F
Frequency	100 kHz to 15 mHz	100 kHz to 10 mHz
AC Current (RMS)	80–130 mA	200 mA

To maintain measurement fidelity, several wiring and instrumentation guidelines were followed:

- All connecting cables were *twisted pair* or braided to minimize electromagnetic noise pickup.

- A *four-wire (Kelvin) connection* was employed for the impedance analyzer leads to eliminate lead and contact resistance from the measurements.
- Current-carrying wires were routed away from voltage-sensing wires to minimize inductive interference.
- The Arbin cycler (used for charging) was electrically isolated from the Solartron EIS instrument during impedance measurements to avoid ground loops or unwanted interactions. In practice, the SC was disconnected from the Arbin and allowed to stabilize at open-circuit potential before the impedance test began.

C. Data Validation and Quality Checks

To ensure the integrity of the impedance data, several validation checks were performed after preliminary EIS measurements and before continuing the main experiment. These validation steps followed established metrological guidelines. Firstly, Open Circuit Potential (OCP) was continuously monitored to verify negligible drift, ensuring that each SC was in a stable state during measurements. Secondly, linearity was confirmed through stable elliptical voltage-current plots, indicating linear and stationary system behavior. Lastly, a linear KK test was employed to validate the impedance data for adherence to causality, linearity, and stability principles. Collectively, this should provide high confidence that the impedance measurements accurately reflect the SCs' behaviour under the tested conditions.

D. Equivalent Circuit Modeling

Correct assessment and characterization of the commercial SCs used in this study requires proper modeling of the porous electrode: electrochemical capacitors with increased performance must exhibit a very high surface area compared to SCs with simple planar electrodes. [15]

The Bisquert Open Circuit model, depicted in Fig. 4, effectively characterizes the porous electrodes of commercial SCs. It employs a transmission line representation that captures the incremental ion transport dynamics within the electrode pores. This schematic illustrates three distinct regions: the electrolyte-base electrode interface, the bulk porous electrode region, and the active interface deeper within the pores. The model comprises parallel and series-connected elements representing electrolyte resistance (R_m), constant phase elements (CPE) for double-layer capacitance, and leakage resistance (R_k) accounting for minimal Faradaic interactions. Such a configuration enables accurate interpretation and analysis through EIS.

The total impedance of a Bisquert Open Circuit model was first calculated by Bisquert [16], [17]. The derivation is based on the location gradient changes of the potential and current within each region. The final impedance is given by:

$$Z_{BTO} = A \coth \left(L \sqrt{R_m \left(\frac{1}{R_k} + Q(i\omega)^\alpha \right)} \right) \quad (2)$$

$$A = \sqrt{\frac{R_k R_m}{1 + Q(i\omega)^\alpha R_k}} \quad (3)$$

with the parameters as indicated in Fig. 4.

E. Fitting Procedure

Impedance data fitting was performed using `impedance.py` [18]. A custom circuit element was implemented based on equation (2), as the Bisquert model is not natively supported. The target fitting accuracy was to be within 5% deviation from the measured data [19].

III. RESULT

The results of the EIS data on the SCs are given in Fig. 6. The All subplots in Fig. 6 exhibit ESR values that remain below the maximum ESR specified in the respective datasheets, indicating reliable and consistent measurements. Moreover, for both SCs, the ESR generally increases with increasing SoC. From literature it is known that the charging process can appreciably deplete the concentration of free ions in the bulk electrolyte within the pores [20]. Both Tables II and III highlight the decrease in ESR with decreasing SoC. Likewise, the diffusion resistance slope seen in Fig. 5a and 5b increases with decreasing SoC below 0.08 Hz and 0.1 Hz respectively. It can also be seen, especially in Fig. 5b, that

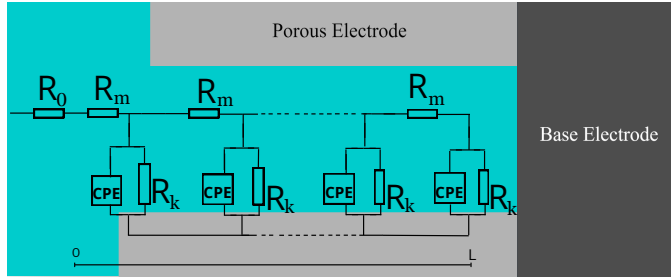


Fig. 4: Bisquert open-circuit ECM with effective pore length L , electrolyte resistance R_m , constant phase elements (CPE), leakage resistance or charge transfer resistance R_k , and equivalent series resistance R_0 .

the impedance curves shift with decreasing SoC. The curves systematically shift toward lower real impedance values as the frequency goes down. This indicates an increase in capacitive dominance at lower SoC. This shift is less visible in Fig. 5a, and one of the reasons might be that the setup is not robust enough to assess a larger SC accurately. Welding the contacts may improve this.

Fig. 6a and Fig. 6b show the results of the Bisquert open circuit model fitting with parameters from Tables III and II, respectively, fitted against the measured impedance data. It can be seen in both Table III and II that R_k is set very high. This is expected with larger EDLC SCs as the parasitic Faradic process plays a much smaller role. This is also evident by the fact that no semi-circle is observed in the EIS plots, which is associated with charge transfer resistance. It is further observed that the capacitance remains largely consistent across both tables, suggesting that the CPE, used to model electrode surface roughness, can be approximated either as an ideal capacitor or as a fixed CPE element. Table II indicates that the pore length decreases as SoC decreases. This interpretation aligns with the theoretical expectation: as the SC charges, more ions become adsorbed onto the porous electrode. The pores thus become deeply filled with ions. Thus, new ions penetrating the layer have to travel further, effectively making the "pore depth" longer [21].

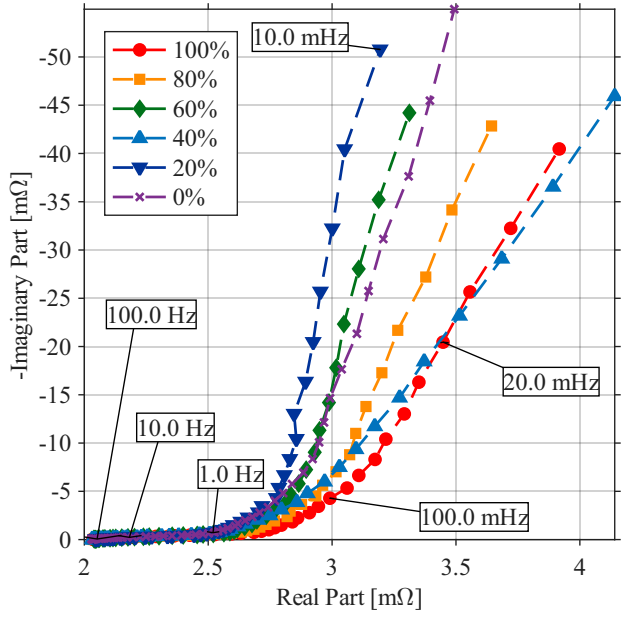
The relative fitting error stays below 5% throughout the spectrum for both SCs except at the transition from the diffusion slope to the capacitive limit, where the relative error increases to around 6%. Table IV provides the root-mean-square error (RMSE) between the fitted and measured vectors. However, the model shows limitations at the transition region between the diffusion-controlled and capacitive-dominated regimes, failing to accurately capture the gentle, concave rise observed in the experimental impedance data seen in both Fig. 6b and 6a.

TABLE II: Bisquert open circuit fit parameters for the 60F SC as a function of SoC.

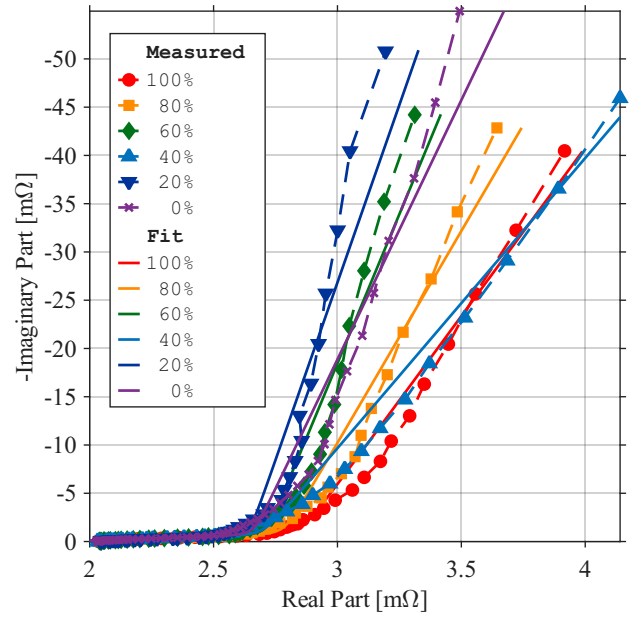
SoC	$R_0[\Omega]$	$R_m[\Omega]$	$R_k[\Omega]$	$Q[\frac{s^\alpha}{\Omega}]$	α	$L[m]$
100%	7.625×10^{-3}	1.527×10^{-2}	1.000×10^{32}	6.034×10^1	9.810×10^{-1}	1.009×10^0
80%	7.481×10^{-3}	1.573×10^{-2}	1.000×10^{32}	6.004×10^1	9.860×10^{-1}	9.179×10^{-1}
60%	7.398×10^{-3}	1.605×10^{-2}	1.000×10^{32}	6.004×10^1	9.864×10^{-1}	8.527×10^{-1}
40%	7.344×10^{-3}	1.720×10^{-2}	1.000×10^{32}	6.004×10^1	9.874×10^{-1}	7.739×10^{-1}
20%	7.379×10^{-3}	1.928×10^{-2}	1.000×10^{32}	5.990×10^1	9.877×10^{-1}	6.732×10^{-1}
0%	7.271×10^{-3}	2.106×10^{-2}	1.000×10^{32}	6.051×10^1	9.906×10^{-1}	6.332×10^{-1}

TABLE III: Bisquert open circuit fit parameters for the 400F SC as a function of SoC.

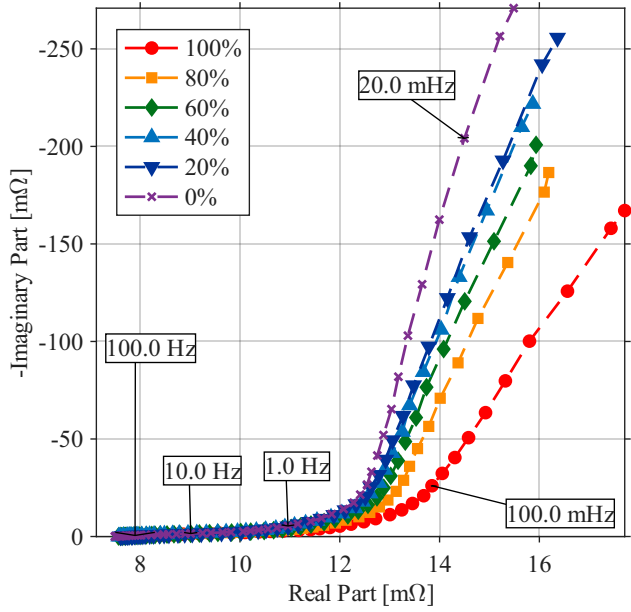
SoC	$R_0[\Omega]$	$R_m[\Omega]$	$R_k[\Omega]$	$Q[\frac{s^\alpha}{\Omega}]$	α	$L[m]$
100%	1.950×10^{-3}	2.925×10^{-3}	2.550×10^3	4.137×10^2	9.818×10^{-1}	9.053×10^{-1}
80%	1.950×10^{-3}	2.848×10^{-3}	2.550×10^3	4.138×10^2	9.855×10^{-1}	8.622×10^{-1}
60%	1.942×10^{-3}	2.696×10^{-3}	2.552×10^3	4.132×10^2	9.897×10^{-1}	8.459×10^{-1}
40%	1.937×10^{-3}	2.798×10^{-3}	2.551×10^3	4.120×10^2	9.789×10^{-1}	7.958×10^{-1}
20%	1.941×10^{-3}	2.831×10^{-3}	2.551×10^3	4.146×10^2	9.914×10^{-1}	7.363×10^{-1}
0%	1.919×10^{-3}	3.237×10^{-3}	2.550×10^3	4.139×10^2	9.882×10^{-1}	6.764×10^{-1}



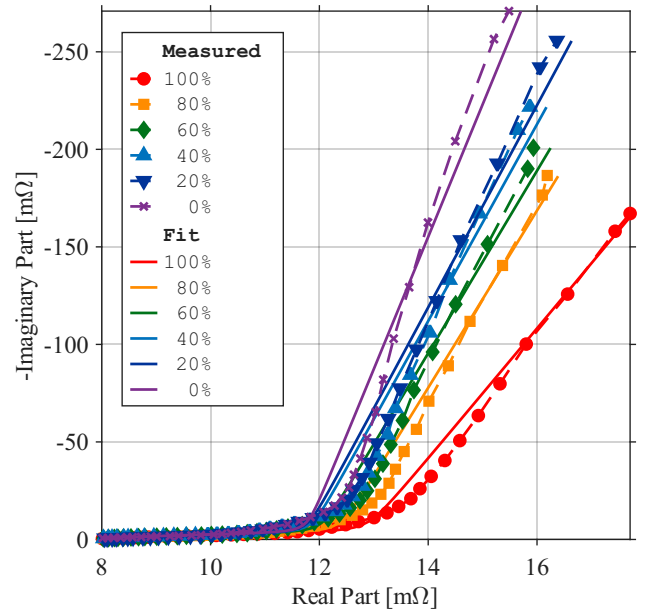
(a)



(a)



(b)



(b)

Fig. 5: Measured Nyquist plot for the 400 F (a) and 60 F (b) supercapacitors using galvanostatic EIS. The frequency decreases from left to right along the graph.

Fig. 6: Fit of the Bisquert open circuit model for the 400 F (a) and the 60 F (b) SC at different SoC Levels.

TABLE IV: RMSE of SCs fits at different SoC levels.

SoC	400 F RMSE [Ω]	60 F RMSE [Ω]
100%	5.481×10^{-5}	2.696×10^{-4}
80%	5.622×10^{-5}	2.774×10^{-4}
60%	6.753×10^{-5}	3.016×10^{-4}
40%	7.696×10^{-5}	3.528×10^{-4}
20%	9.295×10^{-5}	3.668×10^{-4}
0%	8.151×10^{-5}	3.630×10^{-4}

IV. CONCLUSION

A comprehensive model is proposed utilizing manually-extracted EIS features to characterize the SC's empirical and electrochemical behavior as a function of SoC, through parameters derived from the Bisquert open-circuit equivalent circuit model. The method is verified by extensive EIS measurements at various SoCs for two commercially available 60 F and 400 F EDLC devices.

The EIS measurements show impedance characteristics consistent with the expected EDLC behavior, specifically a steep line in the low-frequency region of the Nyquist plot, indicative of capacitive dominance and transitioning toward resistive behavior at higher frequencies. The Bisquert open-circuit model accurately fitted the experimental spectra across all evaluated SoCs, effectively capturing the impedance shifts associated with changes in SoC. At higher SoC, the supercapacitors showed lower ESR and less ideal capacitive behavior in the low-frequency region, indicating enhanced charge availability when the device is less charged. It was also evident that pore length decreases as SoC decreases. Parameter extraction provided valuable insights, including electrolyte resistance (R_m), double-layer capacitance (CPE), and pore lengths (L) as functions of SoC. Across all tested conditions, the model's fitting process achieved high accuracy, with average RMSE errors remaining below 5%, reflecting the model's robustness.

Even better results may be achieved by welding the contacts, especially at higher capacitances. Furthermore, simplifying the model by introducing ideal capacitors instead of CPE and stripping R_k can make the fitting process quicker. This could establish a standard that could be widely adopted due to its reduced complexity, but it would struggle to capture the concave transition region at the diffusion-to-capacitive crossover. Finally, integrating EIS diagnostics such as those presented in this paper with online monitoring or machine learning algorithms is a promising approach to enable real-time SoC and SoH estimation in supercapacitor systems.

REFERENCES

- [1] K. Laadjal and A. J. Marques Cardoso, *A review of supercapacitors modeling, SoH, and SoE estimation methods: Issues and challenges*, *International Journal of Energy Research*, vol. 45, no. 13, pp. 18424–18440, 2021.
- [2] J. Zhao and A. F. Burke, "Review on supercapacitors: Technologies and performance evaluation," *Journal of Energy Chemistry*, vol. 59, pp. 276–291, 2021. doi: [10.1016/j.jchem.2020.11.013](https://doi.org/10.1016/j.jchem.2020.11.013).
- [3] D. Lee, J. Kim, C.-W. Kim, J.-G. Kim, S. E. Jung, S. J. Heo, B. W. Im, N. D. Kim, S. G. Kim, Y. Piao, and B.-C. Ku, "Nanocell-structured carbon nanotube composite fibers for ultrahigh energy and power density supercapacitors," *Composites Part B: Engineering*, vol. 295, Art. no. 112179, 2025. doi: [10.1016/j.compositesb.2025.112179](https://doi.org/10.1016/j.compositesb.2025.112179).
- [4] M. E. Şahin, F. Blaabjerg, and A. Sangwongwanich, "A Comprehensive Review on Supercapacitor Applications and Developments," *Energies*, vol. 15, no. 3, Art. no. 674, 2022, doi: [10.3390/en15030674](https://doi.org/10.3390/en15030674).
- [5] W. Zhang, T. Li, W. Wu, N. Ouyang, and G. Huang, "Data-driven state of health estimation in retired battery based on low and medium-frequency electrochemical impedance spectroscopy," *Measurement*, vol. 211, p. 112597, 2023, doi: [10.1016/j.measurement.2023.112597](https://doi.org/10.1016/j.measurement.2023.112597).
- [6] S. Azizigahlehsari, E. A. Boj, P. Venugopal, T. B. Soeiro, and G. Rietveld, "A Distribution of Relaxation Time Approach on Equivalent Circuit Model Parameterization to Analyse Li-ion Battery Degradation," *IEEE Transactions on Industry Applications*, 2024.
- [7] B. Breazu, S. Azizigahlehsari, P. Venugopal, G. Rietveld, and T. B. Soeiro, "Li-ion Battery Prognostics with Statistical Model and RNN Trained with EIS-Based Features," in *Proc. 2023 IEEE Energy Conversion Congress and Exposition (ECCE)*, pp. 1699–1705, 2023.
- [8] W. Raza, F. Ali, N. Raza, Y. Luo, K.-H. Kim, J. Yang, S. Kumar, A. Mehmood, and E. E. Kwon, "Recent advancements in supercapacitor technology," *Nano Energy*, vol. 52, pp. 441–473, 2018. <https://doi.org/10.1016/j.nanoen.2018.08.013>
- [9] S. Buller, E. Karden, D. Kok, and R. W. De Doncker, "Modeling the dynamic behavior of supercapacitors using impedance spectroscopy," *IEEE Transactions on Industry Applications*, vol. 38, no. 6, pp. 1622–1626, Nov.–Dec. 2002, doi: [10.1109/TIA.2002.804762](https://doi.org/10.1109/TIA.2002.804762).
- [10] M. Catelani *et al.*, "Experimental Characterization of Hybrid Supercapacitor Under Different Operating Conditions Using EIS Measurements," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–10, 2024, Art. no. 3503210, doi: [10.1109/TIM.2023.3329094](https://doi.org/10.1109/TIM.2023.3329094).
- [11] B. E. Conway, *Electrochemical Supercapacitors: Scientific Fundamentals and Technological Applications*. New York, NY: Springer Science & Business Media, 1999, Chapter 18, pp. 555–580.
- [12] Kaus, M., Kowal, J., & Sauer, D. U., "Modelling the effects of charge redistribution during self-discharge of supercapacitors," *Electrochimica Acta*, vol. 55, no. 25, pp. 7516–7523, 2010. [Online]. Available: <https://doi.org/10.1016/j.electacta.2010.01.002>.
- [13] M. Zucca, M. Hassanzadeh, O. Conti, and U. Pogliano, "Accurate parameters identification of a supercapacitor three-branch model," *IEEE Access*, vol. 11, pp. 122387–122398, 2023, doi: [10.1109/ACCESS.2023.3328803](https://doi.org/10.1109/ACCESS.2023.3328803).
- [14] M. A. Zabara *et al.*, "Utility of lissajous plots for electrochemical impedance spectroscopy measurements: Detection of non-linearity and non-stationarity," *Journal of The Electrochemical Society*, vol. 171, no. 1, 010507, Jan. 2024, doi: [10.1149/1945-7111/ad19eb](https://doi.org/10.1149/1945-7111/ad19eb).
- [15] B. E. Conway, *Electrochemical Supercapacitors: Scientific Fundamentals and Technological Applications*, New York, NY: Springer Science & Business Media, 1999, Chapter 6, pp. 110–115, ISBN: 978-0-306-45736-4. doi: [10.1007/978-1-4757-3058-6](https://doi.org/10.1007/978-1-4757-3058-6).
- [16] J. Bisquert, "Theory of the impedance of electron diffusion and recombination in a thin layer," *The Journal of Physical Chemistry B*, vol. 106, no. 2, pp. 325–333, 2002, doi: [10.1021/jp011941g](https://doi.org/10.1021/jp011941g). [Online]. Available: <https://doi.org/10.1021/jp011941g>.
- [17] J. Bisquert, "Influence of the boundaries in the impedance of porous film electrodes," *Physical Chemistry Chemical Physics*, vol. 2, no. 18, pp. 4185–4192, 2000, doi: [10.1039/B001708F](https://doi.org/10.1039/B001708F).
- [18] M. D. Murbach, B. Gerwe, N. Dawson-Elli, and L. K. Tsui, "impedance.py: A Python package for electrochemical impedance analysis," *Journal of Open Source Software*, vol. 5, no. 52, p. 2349, 2020, doi: [10.21105/joss.02349](https://doi.org/10.21105/joss.02349).
- [19] M. Zucca, N. Al-Zubaidi-R-Smith, L. Bartova, H. Brom, L. Callegaro, A. Cultrera, L. Fast, A. Girimonte, M. Hassanzadeh, A. Mariscotti, A. Masouras, J. Medved, S. Musumeci, G. Nicol, M. Ouameur, and G. Rietveld, "MetSuperCap: Metrology for static and dynamic characterisation of supercapacitors," *Measurement: Sensors*, 2024, Art. no. 101434, doi: [10.1016/j.measen.2024.101434](https://doi.org/10.1016/j.measen.2024.101434).
- [20] B. E. Conway, *Electrochemical Supercapacitors: Scientific Fundamentals and Technological Applications*. New York, NY: Springer Science & Business Media, 1999, Chapter 17, pp. 525–538.
- [21] Krishnan, S. G., Arulraj, A., Jagadish, P., Khalid, M., Nasrollahzadeh, M., Fen, R., ... Hegde, G. (2022). Pore size matters!—a critical review on the supercapacitive charge storage enhancement of biocarbonaceous materials. *Critical Reviews in Solid State and Materials Sciences*, 48(1), 1–56. <https://doi.org/10.1080/10408436.2022.2027225>